



UNSW
SYDNEY

School of Mechanical and Manufacturing Engineering
Faculty of Engineering
UNSW Sydney

Aidan Knipe

z5418749

**Macro-Micro Robotic System with a
Flexible Continuum Tip for Minimally
Invasive Surgery**

Interim Report and Project Plan

Abstract

This draft literature review is structured for a thesis on a UR5e-based macro-micro robotic system with a flexible continuum tip for minimally invasive surgery (MIS). It consolidates relevant findings from prior work in the *Lachlan Report*, identifies which foundations remain valid, and highlights where the state of the art now requires updated evidence. The review prioritizes three technical threads central to this thesis: (1) tendon-driven continuum manipulator design and motion algorithms, (2) integration with a macro robot (UR5e) with improved teleoperation behaviour, and (3) rapid end-effector exchange mechanisms. It also includes explicit “missing area” headings to direct next-stage searching, reading, and synthesis.

ORIGINALITY STATEMENT

‘I hereby declare that this submission is my own work and to the best of my knowledge it contains no materials previously published or written by another person, or substantial proportions of material which have been accepted for the award of any other degree or diploma at UNSW or any other educational institution, except where due acknowledgement is made in the thesis. Any contribution made to the research by others, with whom I have worked at UNSW or elsewhere, is explicitly acknowledged in the thesis. I also declare that the intellectual content of this thesis is the product of my own work, except to the extent that assistance from others in the project’s design and conception or in style, presentation and linguistic expression is acknowledged.’

Signed: _____

Date: _____

Contents

Originality Statement	ii
List of Figures	v
List of Tables	vi
Nomenclature	vii
I Introduction	1
I.1 Project Context and Problem Significance	1
I.2 Working Thesis Gap Statement (Draft)	1
II Literature Review	1
II.1 Introduction to Minimally Invasive Surgery (MIS) and Robotics	1
II.2 Teleoperation and Control in Surgical Robotics	4
II.2.1 Modelling Foundations Relevant to This Thesis	4
II.3 Modularity and Tendon-Driven Continuum Manipulators	8
II.3.1 Tool Exchange Mechanisms	8
II.4 System Safety and Reliability Considerations	10
II.4.1 3.1 Safety-Critical Hazard Model for Macro-Micro MIS	10
II.4.2 3.2 Distal Contact and Force-Limiting Strategies	10
II.4.3 3.3 Teleoperation Reliability Under Delay, Jitter, and Tremor	10
II.4.4 3.4 Backlash/Hysteresis and Compensation Stability	11
II.4.5 3.5 Tool-Exchange Safety Recovery	11
II.4.6 3.6 Fault Handling and Safe-State Behavior	11
II.4.7 3.7 Reliability Evidence Across Lifecycle	11
II.4.8 3.8 Section-Level Critical Synthesis and Gap Statement	12
II.5 Synthesis of Literature and Knowledge Gaps	12
II.6 Priority Evidence Gaps for Additional Review	13
III Research Question and Project Plan	13
III.1 Research Question and Plan Status	13
III.2 Plan Quality Criteria (Assessment-Aligned)	13
III.3 Current Planning Position	14
IV Project Dependent Preparations	14
V Conclusion	14
Acknowledgements	A

References	B
Appendices	D
A Rubric-Aligned Coverage Notes (MMAN4951/MMAN9451 Interim Report)	D

List of Figures

1	Progression of surgery and enabling technology from conventional approaches to modern MIS robotics.	2
2	SnakeRaven macro-micro surgical manipulator concept used as a relevant continuum-tip reference from prior work (Scott, 2022; Razjigaev et al., 2021).	4

List of Tables

Nomenclature

(The nomenclature must respect the following order: latin symbols, Greek symbols, acronyms. Within each category of symbols, follow alphabetic order. An example is provided below)

A_c	=	tendon/cable cross-sectional area
D	=	backlash deadband
E	=	Young's modulus of tendon/cable material
M_{grasp}	=	estimated distal grasping torque
R_0	=	driving pulley radius
R_5	=	driven pulley radius
R_{eq}	=	equivalent-circle radius for tendon-sheath geometry
R_i	=	orientation matrix of segment i
T_i	=	homogeneous transform of segment i
$\Delta\theta_{out}$	=	output-side angular backlash/displacement loss
Ω	=	total cumulated curve angle of a tendon-sheath path
θ_{comp}	=	feedforward backlash compensation term
μ	=	friction coefficient (cable-pulley contact)
θ_i	=	local bending angle of segment i
α_i	=	local orientation offset of segment i
$\mathbf{p}$	=	rod centerline position in Cosserat formulation
$\mathbf{R}$	=	rod orientation in Cosserat formulation
$\mathbf{n}$	=	internal force in Cosserat formulation
$\mathbf{m}$	=	internal moment in Cosserat formulation
C_p	=	...
C_x	=	...
C_y	=	...
α	=	...
MIS	=	minimally invasive surgery
TLA	=	...

I Introduction

I.1 Project Context and Problem Significance

This thesis extends prior macro-micro robotic surgery work by coupling a UR5e macro manipulator with a new flexible continuum tip, then improving movement algorithms inherited from earlier implementations. The applied significance is strong: MIS demands dexterity in constrained spaces, low tissue interaction force, high positional repeatability, and robust safety controls.

From the earlier project, the macro-micro concept and teleoperation feasibility are already established at a proof-of-concept level. The key opportunity now is to move from demonstration toward a more clinically credible engineering argument through stronger modeling, improved control reliability, and modular hardware architecture (rapid tool exchange).

I.2 Working Thesis Gap Statement (Draft)

Current literature and prior project work support feasibility of macro-micro MIS manipulation, but leave a practical translation gap at the intersection of **accurate continuum movement control**, **stable and reliable teleoperation**, and **fast, repeatable end-effector reconfiguration**. This thesis addresses that gap on a UR5e-based platform by improving inherited motion algorithms and integrating reliability and modularity mechanisms necessary for repeatable experimental deployment.

II Literature Review

II.1 Introduction to Minimally Invasive Surgery (MIS) and Robotics

Laparoscopic surgery is a core MIS approach in which instruments are inserted through small abdominal incisions using trocar access ports, rather than a large open incision (Ko et al., 2025). In this report, MIS denotes procedures that use these reduced-access pathways to lower tissue trauma and recovery burden relative to open surgery.

- The clinical motivation: Why do we need surgical robots? (Precision, dexterity, patient outcomes).
- The rise of Macro-Micro systems (combining a large positioning arm like a UR5e with a delicate continuum tip).

Historical progression of surgery and MIS robotics Technology and surgery have been inseparable for centuries as every surgical advancement or care improvement has been enabled by the parallel development of instruments and tools. From anaesthesia to antiseptics, innovation has enabled surgeons to reduce risk and expand the scope of treatable conditions (Emerging Robotic Platforms for Minimally Invasive Surgery) In the 19th century, innovations such as the transition from candlelight to electric lighting allowed for practical internal visualisation, marking the beginning of a major paradigm shift in the surgical field (Hargest five thousand years). Electricity also enable the development of various specialised surgical procedures, such as electrocautery for haemostatis and the use of electric drills for orthopaedic surgery (hargest 5000 years). These innovations allowed for a stronger focus on precision and dexterity, and techniques such as laparoscopy and endoscopy were developed at the beginning of the 20th century, resulting in some of the first minimally invasive surgeries (MIS)(Hatzinger M, Fesenko A, Sohn M. The first human laparoscopy and NOTES operation: Dimitrij Oscarovic Ott (1855-1929). Urol Int. 2014;92(4):387-91. doi: 10.1159/000358016. Epub 2014 May 20. PMID: 24852454.). This trend towards less invasive procedures has continued, with the development of revolutionary robotic surgical systems such as the da Vinci Surgical System in the late 1990s, which has become widely adopted for various surgical procedures (Intuitive Surgical. da Vinci Surgical System. <https://www.intuitive.com/en-us/products-and-services/da-vinci/surgical-systems>. Accessed June 2024).

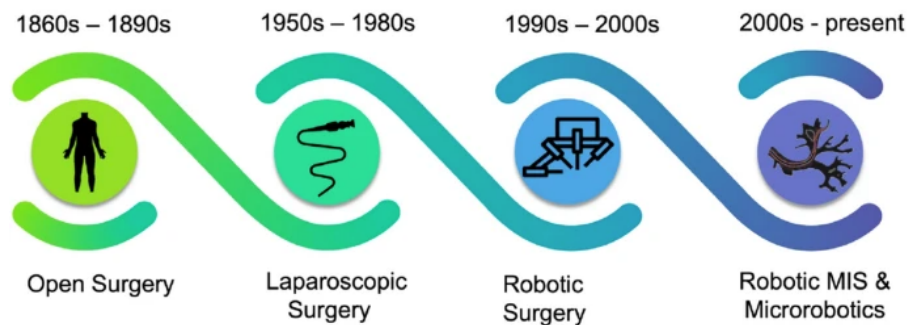


Figure 1: Progression of surgery and enabling technology from conventional approaches to modern MIS robotics.

Macro-micro architecture motivation The modern frontier of surgical robotics is achieving human-level dexterity, accuracy, and force control in increasingly small and complex anatomical spaces. One key technological direction enabling this transition is minimally invasive surgical tooling with a macro-micro architecture. The macro unit provides large-workspace positioning and efficiency; however, its larger structure can be prone to deformation and vibration. The micro manipulation unit is then attached to the larger arm and provides the distal dexterity required for manipulation in confined anatomy, where surgical targets often require end-effectors in the 3–4 mm class for practi-

cal access through narrow channels and natural orifices. In current snake-robot literature, this micro stage is typically tendon/cable-actuated to achieve high local curvature and fine orientation control, but its performance depends strongly on motion transmission quality, sensing, and control architecture (Seetohul and Shafiee, 2022). In a macro-micro arrangement, this local dexterity complements the macro arm’s gross positioning capability and helps isolate the endpoint from macro-stage vibration through coordinated control (Scott, 2022).

Cost and accessibility context With growing demand and a rapidly expanding surgical robotics market, there is ongoing debate over whether the high cost of integrated platforms such as da Vinci limits accessibility, with acquisition costs reported up to \$4 million AUD and per-procedure costs of roughly \$1000–\$5000 (Do the costs of robotic surgery present an insurmountable obstacle? A narrative review). Despite clear clinical benefits, providers must still evaluate cost-effectiveness, especially in resource-limited settings (Do the costs of robotic surgery present an insurmountable obstacle? A narrative review). Evidence from low-to-middle income settings further shows that adoption barriers are not limited to purchase price alone: recurring maintenance, disposable instrument costs, and sustained training requirements can prevent programme continuity even after initial installation (Raza et al., 2024). The same review also highlights system-level constraints, including limited infrastructure, workforce training gaps, and tension between investing in robotics versus addressing unmet basic surgical needs, reinforcing a clear translational gap for scalable low-cost platforms (Raza et al., 2024). Taken together, these findings indicate a specific gap: affordable macro-micro systems that combine low-cost hardware with delay-robust, clinically credible teleoperation performance are still underdeveloped. These findings support the broader literature position that low-cost macro-micro architectures, including off-the-shelf hardware and open-source software stacks, are an important direction for improving accessibility while maintaining MIS precision and safety requirements (Scott, 2022).

UR5e baseline and observed limitation Specifically, the UR5e is a widely used, cost-effective robotic arm that can serve as the macro stage, with reported pricing around \$50,000–\$60,000 AUD, and can be integrated with a range of custom micro systems (Universal Robots price guide: What to expect (new and used costs)). The use of a UR5e-based macro-micro architecture has been demonstrated in prior thesis work by L. Scott (2022), which provided a proof-of-concept teleoperation pipeline using open-source ROS2. However, as highlighted by Scott, translational macro-arm commands produced observable vibration in the integrated system. Reducing this behaviour is necessary to meet clinical expectations for safety, repeatability, and motion quality in operating environments.

Continuum tips are a clinically relevant class of hyper-redundant end-effectors for

MIS because they combine high dexterity with slender geometries that can be advanced through confined access pathways (e.g., lumens, natural orifices, and small incisions), where conventional rigid instruments are mechanically constrained. In the tendon-driven subclass, curvature is generated through differential tendon tension about a central backbone or articulated serial structure, enabling multi-DOF distal steering with compact proximal actuation and making these devices well suited to macro-micro architectures (Scott, 2022; Razjigaev et al., 2021; Seetohul and Shafiee, 2022).

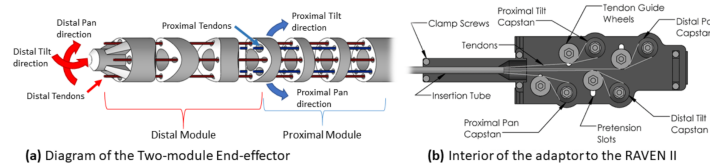


Figure 2: SnakeRaven macro-micro surgical manipulator concept used as a relevant continuum-tip reference from prior work (Scott, 2022; Razjigaev et al., 2021).

Within this context, current UR5e-based continuum implementations in the local literature follow the tendon-driven snake-like micro-manipulator direction established in Lachlan Scott’s thesis, including a 4 mm-class distal tool, macro-micro integration with a UR5e platform, and teleoperation via a haptic interface. Scott reports preliminary electromagnetic-tracking validation showing close pose mirroring under teleoperation (reported average absolute position errors of 0.0272 mm and 0.0161 mm across trial groups), while also identifying practical engineering constraints that remain unresolved, including macro-stage translational vibration and the time-consuming exchange of tendon-driven end-effectors in the current modular workflow (Scott, 2022; Razjigaev et al., 2021; Razjigaev et al., 2022; Seetohul and Shafiee, 2022).

II.2 Teleoperation and Control in Surgical Robotics

- *The Problem:* Human tremor, system latency, and mapping user intent to rigid vs. continuum kinematics.
- *Current Solutions:* Review papers on teleoperation filters (Kalman filters, moving average, predictive kinematics).
- *The Gap:* Discuss how high-end systems do this well, but open-source/low-cost systems (like Lachlan’s) suffer from vibratory feedback across ROS/external control nodes.

II.2.1 Modelling Foundations Relevant to This Thesis

To situate control and modularity claims on a consistent mathematical basis, three modelling layers from continuum-robotics literature are directly relevant to this project.

At the kinematic level, each segment transform can be written as

$$T_i = \begin{bmatrix} R_i & p_i \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad (1)$$

which supports forward mapping from joint/segment states to distal pose and is the baseline representation used across continuum and hyper-redundant modelling frameworks (Burgner-Kahrs et al., 2015; Dupont et al., 2022).

For tendon-driven discrete continuum segments, piecewise-constant-curvature formulations remain practical for actuator-to-workspace mapping:

$$T_i^{i+1} = \text{rot}(Z_i, \alpha_i) \text{trans}(Z_i, H_i) \text{trans}(X_i, W_i) \text{rot}(Y_i, \theta_i) \text{rot}(Z_i, -\alpha_i), \quad (2)$$

which is the core structure behind recent high-DOF tendon-driven modelling and calibration workflows (Wang et al., 2022; Dupont et al., 2022).

When compliance, external loading, and routing-dependent behavior become dominant, Cosserat-rod equilibrium/dynamics provide the stronger foundation:

$$\mathbf{p}_s = \mathbf{R}\mathbf{v}, \quad \mathbf{R}_s = \mathbf{R}\hat{\mathbf{u}}, \quad \mathbf{n}_s = \rho A \mathbf{R}(\hat{\boldsymbol{\omega}}\mathbf{q} + \mathbf{q}_t) - \mathbf{f}, \quad \mathbf{m}_s = (\rho \mathbf{R}\mathbf{J}\boldsymbol{\omega})_t - \hat{\mathbf{p}}_s \mathbf{n} - \mathbf{l}. \quad (3)$$

This class of models is especially relevant to this thesis where tendon friction/hysteresis, backlash compensation, and post-swap recalibration are coupled through compliant routing effects rather than purely rigid kinematics (Janabi-Sharifi et al., 2021).

In critical terms, the literature indicates that no single layer is sufficient for workflow-ready surgical performance: kinematic models are computationally efficient but can under-represent load-dependent effects, while full mechanics-based models are more expressive but harder to identify and maintain in real time. This trade-off directly motivates hybrid compensation strategies in the subsequent teleoperation and modularity sections.

Within this control context, the relevant evidence can be read in five linked layers: tremor suppression, latency compensation, robust state estimation, predictive bilateral control, and intent-aware shared autonomy.

Human physiological tremor is a central control challenge in robot-assisted MIS because involuntary operator motion is transmitted through the teleoperation loop and can appear as vibration at the surgical instrument tip, degrading precision during fine tissue manipulation, suturing, and needle handling. Sang et al. (2016) report that tremor suppression methods must balance two competing requirements: strong attenuation of involuntary motion and minimal phase delay, as latency in the command pathway can directly

reduce teleoperation quality. Their zero-phase adaptive fuzzy Kalman filter (ZPAFKF) combines a zero-phase prefilter with adaptive Kalman estimation to improve tremor estimation and compensation in real time, and experimental master-slave results show lower error and reduced tip vibration relative to standard filtering baselines. This is directly relevant to the present thesis because it frames tremor control not as a standalone signal-processing problem, but as a coupled teleoperation and tracking problem in which compensation quality, delay, and manipulator dynamics must be considered together (Sang et al., 2016).

System latency is a second critical bottleneck in bilateral teleoperation, particularly when haptic feedback is included. Batty et al. (2022) show that communication delay creates a fundamental transparency-stability trade-off: increasing direct force correspondence can degrade stability, while aggressive stabilisation can mute the fidelity of operator feedback. In response, they implement an environment-estimation and force-prediction architecture, where slave-side contact parameters are estimated online (EWRLS) and master feedback is generated through a predicted virtual environment rather than delayed direct force reflection. Their human-in-the-loop palpation experiments demonstrate that this predictive approach substantially reduces force prediction error (with the Hunt-Crossley model outperforming Kelvin-Voigt and direct force feedback), and improves teleoperation transparency under delay. A key implication in the literature is that latency management should be treated as a control-architecture problem, not solely a communications problem, and that model quality strongly influences the practical quality of teleoperated manipulation (Batty et al., 2022).

Filtering is therefore not optional in surgical teleoperation; it is a foundational requirement for maintaining stable and trustworthy state estimates when measurements are noisy and network transport is imperfect. Lashari et al. (2025) show that a Kalman filter paired with MOESP-identified state-space models can estimate patient-side manipulator position with high robustness under delay, jitter, and packet loss, while remaining computationally lightweight for real-time use. This is especially relevant when exact plant dynamics are not available a priori, because MOESP builds the model directly from surgical kinematic data (JIGSAWS), allowing filtering performance to be grounded in observed system behaviour rather than idealised assumptions. In practical terms, filters are needed to smooth stochastic disturbances, reconstruct missing or delayed information, and preserve control continuity so that downstream motion control and haptic/visual feedback remain clinically usable. A key limitation, however, is that the reported performance is primarily based on dataset-driven modelling and simulated network impairments, so in vivo generalisability and cross-platform transfer still require further validation (Lashari et al., 2025).

Predictive bilateral control is also emerging as a practical strategy for continuum and snake-like teleoperation under delay. Ebrahimian et al. (2024) propose an online architec-

ture that combines model-mediated force prediction on the master side with ANN-based operator motion prediction on the slave side, with the explicit goal of reducing perceived communication lag while preserving position synchronisation and haptic transparency. This is highly relevant to the macro-micro literature because it addresses the same core bottleneck as latency and filtering studies, but at the controller-architecture level: delay compensation is handled through coordinated forward and backward prediction, rather than through one-sided compensation only. Their simulation results show strong delay bypass for known constant delays in snake-like robot teleoperation, but a key limitation is that performance is demonstrated mainly in simulation with structured assumptions (e.g., known constant delay and constrained robot DOFs), which may not fully capture variable-delay clinical networks or higher-complexity surgical interaction dynamics (Ebrahimian et al., 2024).

Beyond tremor and delay, user intent mapping is a central teleoperation design problem in shared control. Bowman et al. (2024) argue that direct motion mapping alone is insufficient for robust telemanipulation when the human and robot hands are physically dissimilar, because low-level geometric tracking does not necessarily preserve high-level task success. Their intent-based task-oriented shared control framework instead blends operator-following behaviour with autonomous task assistance using inferred intent probabilities and adaptive arbitration weights. Critically, this contribution is not just another blending policy: it formalises how authority should shift based on hand-structure discrepancy and task constraints, which is directly relevant to macro-micro MIS systems where distal tools cannot perfectly mimic surgeon hand kinematics. This supports the broader literature position that intuitive teleoperation should be evaluated by both task completion and perceived control, rather than endpoint tracking error alone (Bowman et al., 2024).

However, an important limitation is that the reported validation is constrained to controlled grasping tasks and specific intent models, so transferability to highly dynamic, deformable, and safety-critical in vivo surgical contexts remains an open question.

Transmission backlash in cable-pulley drives is another directly relevant control bottleneck for laparoscopic end-effectors because direction reversals create delay and transition phases in which distal motion/torque do not track proximal commands linearly. Zou et al. (2022) model this explicitly and use a feedforward compensation term linked to output-side backlash, $\theta_{comp} = \Delta\theta_{out} R_5/R_0$, together with grasping torque estimation and saturation limiting, $M_{grasp} = M_{out} - M_{fric}$, to reduce tracking error and limit unsafe distal force. Their results indicate meaningful improvements in position and force control for cable-pulley laparoscopic instruments: reported distal-grasper position error is reduced to approximately 1.03° mean (3.49° max), and grasping torque tracking error is reported below $0.025 \text{ N}\cdot\text{m}$ in their compensated instrument tests. Nonetheless, residual error around direction reversal remains, highlighting that compensation quality still

depends on parameter identification and friction-model fidelity.

Related tendon–sheath evidence further strengthens this point in shape-variable conditions. Hong et al. (2025) model backlash-like hysteresis in tendon–sheath pairs using an equivalent-circle representation, where deadband is linked to master/slave deformation mismatch ($D = \delta_{TSM(master)} - \delta_{TSM(slave)}$) and then compensated by feedforward updated from proximal calibration without distal sensing. Their reported results show RMSE reduction from 1.31 mm to 0.19 mm on average (85.5% reduction). This is highly relevant to laparoscopic routing uncertainty; however, the approach still depends on reliable calibration and assumptions about unchanged geometric conditions during each compensation interval.

Collectively, these control studies establish clear performance requirements for low-cost macro-micro platforms and provide a direct conceptual bridge to modular continuum hardware discussions.

II.3 Modularity and Tendon-Driven Continuum Manipulators

In this review, a *monolithic* distal architecture is defined as a single integrated tool assembly in which changing function requires major teardown or full distal module replacement rather than rapid local exchange. This framing is consistent with the architecture-level distinction between integrated and modular robotic platforms reported by Cowan et al. (2024).

The practical consequences are workflow-critical: longer swap time, re-tensioning and re-calibration burden after replacement, interruption of the operative sequence, and reduced intraoperative adaptability when task needs change mid-procedure. These constraints align with broader integration-friction evidence in modular end-effector software/hardware pipelines (Torielli et al., 2023), where even technically feasible modularity often carries non-trivial changeover overhead.

Internal platform observations in the prior local UR5e-continuum implementation also indicate that end-effector replacement is time-consuming and introduces practical workflow friction around repeatable setup after each swap (Scott, 2022). Together, the literature and local prior-work evidence indicate that architectural modularity alone is insufficient unless exchange-time and post-swap recalibration costs are explicitly addressed.

II.3.1 Tool Exchange Mechanisms

Tool exchange is necessary because laparoscopic workflows require multiple functional instruments within one procedure, while only a limited subset can be mounted simultaneously on the robotic arm (Ko et al., 2025). This creates a recurrent intraoperative transition task that can interrupt surgical flow and, when assistant-dependent, introduces operator-assistant coordination variance.

At present, the strongest direct evidence remains with rail-guided, cartridge-like exchange approaches. Ko et al. (2025) analyse current sliding exchange practice and report an automated replacement architecture designed to preserve trocar/RCM constraints via linear motion. Their prototype-level results report full exchange in 15 s, reduction from an initial 27 s cycle, and a total device mass of 3.36 kg (below their 4 kg design target), indicating feasible automation under laparoscopic geometric constraints.

Alternative strategies aim to reduce exchange frequency rather than improve exchange interfaces directly. Mocellin et al. (2026) present OriGrasp as a reconfigurable multifunctional end-effector (introducer mode and compliant grasper mode), validated through dVRK integration and tissue-interaction tests. Reported performance includes approximately 4 N pinch force and bowel pulling/ lifting forces in literature-consistent atraumatic ranges. This supports multifunctionality as a useful partial mitigation for selected abdominal tasks. However, current evidence remains task-specific and does not demonstrate that a single reconfigurable device can replace the broader instrument set required across heterogeneous procedures.

Mechanism diversity beyond rail-guided exchange remains under-validated in the reviewed evidence base. Direct surgical evidence is limited for bayonet quarter-turn, cam-latch, cartridge-head variants beyond linear sliding, and kinematic-coupling-assisted docking with active preload. Reported gaps are not purely mechanical: tolerance sensitivity, sterilisation robustness, wear-dependent repeatability drift, and post-reconnection re-zeroing burden all remain insufficiently quantified across repeated swap cycles.

Control evidence indicates that docking success alone is insufficient for workflow-ready exchange. Cable-pulley and tendon-sheath studies report backlash/hysteresis and force-estimation sensitivity to pretension, friction, and routing geometry (Zou et al., 2022; Hong et al., 2025). In particular, Hong et al. (2025) show geometry-dependent deadband behavior and report RMSE reduction from 1.31 mm to 0.19 mm with feedforward compensation after calibration, implying that post-swap recovery may require both tension re-establishment and compensation-state updates when routing changes.

For post-swap tension management specifically, Singh and Khatait (2022) provide relevant but indirect evidence via an RCM tension-adjustment mechanism enabling independent cable pretension tuning with invariant routed tendon length. The key limitation is translational scope: this is not a direct quick-change end-effector interface study and does not quantify repeated dock-undock surgical workflow performance.

Critical judgement and knowledge gap. The literature supports a clear hierarchy of maturity: (i) strongest evidence for linear, rail-guided exchange, (ii) promising but task-limited multifunctional substitution, and (iii) weak comparative evidence for alternative docking/locking families. The unresolved research gap is therefore the integrated validation of *mechanism + control recovery* after repeated swaps: exchange time, docking repeatability, post-swap backlash/deadband, retensioning effort, and force-safety

consistency must be quantified together under clinically realistic routing and sterilisation conditions.

II.4 System Safety and Reliability Considerations

This section is structured as a staged argument from hazard definition to quantified residual risk, so that the final gap is explicit and testable.

II.4.1 3.1 Safety-Critical Hazard Model for Macro-Micro MIS

Purpose: Define what can go wrong and what must be bounded.

- enumerate hazard classes: unintended contact, excessive distal force, delayed response, post-swap miscalibration, and control-state mismatch after direction reversal,
- state measurable safety variables: distal force/torque limits, position error, response latency, and allowed deadband/backlash window,
- link each hazard to literature-supported mechanism/control causes.

II.4.2 3.2 Distal Contact and Force-Limiting Strategies

Purpose: Show how current work constrains tissue interaction risk.

- compare force-limiting approaches: intrinsic compliance, model-based force estimation, and torque saturation,
- extract quantitative thresholds/results where available (e.g., force-tracking and grasp-safety outcomes),
- identify missing evidence: transfer from benchtop/tissue-phantom performance to clinically variable in vivo conditions.

II.4.3 3.3 Teleoperation Reliability Under Delay, Jitter, and Tremor

Purpose: Establish reliability limits of the command loop itself.

- synthesise predictive control, filtering, and intent arbitration literature as complementary layers,
- assess robustness to delay and packet-loss assumptions versus real operating-network variability,
- identify unresolved issue: consistent safety-performance under simultaneous delay, model mismatch, and human adaptation.

II.4.4 3.4 Backlash/Hysteresis and Compensation Stability

Purpose: Connect transmission nonlinearity directly to safety and repeatability.

- review backlash-like hysteresis models for cable-pulley and tendon-sheath paths,
- summarise compensation mathematics and reported residual errors,
- identify reliability gap at direction reversals and post-calibration drift.

II.4.5 3.5 Tool-Exchange Safety Recovery

Purpose: Treat swap events as safety-critical state transitions, not only mechanical operations.

- define post-swap recovery requirements: tension re-establishment, deadband update, re-zeroing, and force-limit consistency,
- evaluate what exchange studies quantify versus what they omit (repeated-cycle degradation, sterilisation-driven wear, and cumulative calibration burden),
- identify coupling gap: mechanism docking success does not guarantee control-state safety restoration.

II.4.6 3.6 Fault Handling and Safe-State Behavior

Purpose: Evaluate how systems fail, not only how they track.

- synthesise evidence on safe-stop logic, bounded fallback behavior, and operator override pathways,
- map expected failure modes (sensor dropouts, cable slack, communication interruption) to mitigation strategies,
- identify missing quantitative benchmarks for recovery time and residual error after fault events.

II.4.7 3.7 Reliability Evidence Across Lifecycle

Purpose: Determine whether reported performance is durable.

- review evidence for repeatability across prolonged trials, repeated swaps, and changing pretension states,
- include sterilisation/contamination and wear effects where available,
- identify the dominant gap: limited long-horizon, clinically representative reliability datasets for low-cost macro-micro platforms.

II.4.8 3.8 Section-Level Critical Synthesis and Gap Statement

Purpose: Conclude with a single defensible knowledge gap.

- synthesise that current literature validates isolated safety mechanisms but rarely validates integrated safety recovery after reconfiguration,
- state thesis-relevant gap: jointly validated *mechanism + control + recovery* safety performance under realistic routing, delay, and swap conditions,
- define candidate evaluation endpoints for later chapters: post-swap recovery time, residual deadband/backlash, distal-force limit adherence, and task-level repeatability.

II.5 Synthesis of Literature and Knowledge Gaps

Across recent literature, there is strong agreement that high-quality surgical teleoperation depends on coordinated management of tremor, delay, state-estimation uncertainty, and operator intent. However, most validated architectures are demonstrated either on high-cost integrated systems or under simplifying assumptions (e.g., known constant delay, controlled tasks, or simulation-only evaluation), which limits generalisability to resource-constrained and clinically variable settings.

In parallel, continuum and snake-like end-effectors provide clear distal dexterity advantages for confined MIS workspaces, but persistent control difficulties remain around tendon friction, hysteresis, backlash, calibration drift, and repeatability after hardware reconfiguration. Evidence from low-to-middle income contexts also shows that long-term adoption depends not only on capital cost but on maintenance, disposables, training continuity, and infrastructure support.

Taken together, the most consequential knowledge gaps are:

- integrated control frameworks that retain transparency and stability under realistic communication imperfections,
- reproducible evaluation protocols for low-cost macro-micro systems using clinically meaningful performance metrics,
- robust, repeatable modular continuum interfaces that minimise post-swap uncertainty,
- long-horizon reliability evidence (wear, sterilisation, recalibration burden) for tendon-driven reconfigurable tools.

II.6 Priority Evidence Gaps for Additional Review

The review can be strengthened further by adding recent studies on:

- benchmark datasets and standardised test protocols for continuum surgical teleoperation,
- variable-delay and packet-loss robust bilateral/shared control in human-in-the-loop settings,
- safety assurance frameworks (fault handling, bounded motion, force-limited behaviour) for MIS robots,
- quantitative studies of modular tool-exchange repeatability and calibration drift,
- lifecycle reliability of tendon-driven instruments under repeated use and sterilisation.

III Research Question and Project Plan

III.1 Research Question and Plan Status

This section is maintained as a separate planning section that draws on the literature review but is not part of the literature-review narrative itself. At this stage, the research question and plan are presented with sufficient detail to indicate feasibility, expected resource requirements, and realistic time demands.

III.2 Plan Quality Criteria (Assessment-Aligned)

The planning section is intended to demonstrate the following progression:

- The research question and plan are clearly presented with enough detail to support feasibility and basic thesis structure.
- The plan is explicitly informed by the reviewed literature, with a clear explanation of why each major work package follows from identified evidence gaps.
- The plan includes realistic milestones, a logical thesis outline, and direct links between chapter flow, reviewed evidence, and planned validation.
- The plan includes contingencies for project variation (e.g., hardware delays, integration issues, data-quality limitations) and alternative execution pathways.

III.3 Current Planning Position

The current project plan is retained as a standalone section for subsequent development. The immediate objective is to preserve alignment with the literature-derived gaps while deferring detailed milestone refinement to the next planning pass.

IV Project Dependent Preparations

All required project-dependent preparations should be tracked here and reviewed for readiness before execution. This section should confirm whether preparations are complete and whether significant revision is still required. It should also record exceptional early progress where available, so that planning assumptions can be confirmed or adjusted and preliminary answers to research questions can be documented.

V Conclusion

The reviewed literature shows a clear convergence on three points. First, macro-micro architectures and tendon-driven continuum tools are well motivated for MIS dexterity, but remain sensitive to vibration transmission, modelling error, and teleoperation instability. Second, modern control strategies (filtering, latency compensation, predictive bilateral control, and intent-aware shared autonomy) improve performance, yet are often validated under constrained assumptions that limit direct transfer to clinically variable settings. Third, cost and adoption studies indicate that accessibility is governed by full-system burden (maintenance, disposables, training continuity, and infrastructure), not capital cost alone.

Taken together, the strongest unresolved knowledge gaps are integrated, reproducible frameworks that jointly address delay-robust control, modular reconfiguration repeatability, safety-critical reliability, and realistic cost-constrained deployment conditions.

Acknowledgements

I would like to thank...

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A Rubric-Aligned Coverage Notes (MMAN4951/MMAN9451 Interim Report)

Criterion 1: Literature Review (High Weighting)

To target Distinction/High Distinction bands, this review must do more than summarize papers. It should:

- Show conceptual links between manipulator design, control, safety, and modularity.
- Explicitly identify and justify knowledge gaps your thesis addresses.
- Include recent work (not only foundational papers).
- Critically compare methods instead of listing them.

Criterion 4: Document Presentation

For marks protection, maintain:

- consistent citation formatting,
- clear figure/table captions,
- polished grammar and structured argument flow.